

DEFINITION-LEVEL COUNTEREXAMPLE

A Rank-One Counterexample to the Literal Form of AIM Problem 1.44

Every nonempty shell of $a\mathbb{Z}$ is the full zero-sphere and hence a spherical t -design for every t

$$\begin{array}{lll} \text{rank 1} & \text{every nonempty shell} & \text{all strengths } t \\ L = a\mathbb{Z}, & L_{a^2k^2} = \{-ak, ak\}, & \frac{1}{ak} L_{a^2k^2} = S^0 \end{array}$$

Scope boundary. This note refutes only the dimension-unqualified sentence printed as AIM Problem 1.44. It does not settle a modified problem imposing $n \geq 2$ or a narrower lattice class. No claim of worldwide priority is made.

中文结论

AIM Problem 1.44 的正式文本没有维数限制。对任意 $a > 0$ ，满一维格 $a\mathbb{Z}$ 的每个非空正范数壳层归一化后就是整个零维球面 $S^0 = \{-1, 1\}$ ，所以对任意强度 t 都是球面 t -设计，特别是球面 12-设计。该结论不涉及另加 $n \geq 2$ 的修改版本。

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Independent audit complete

Contents

1 Abstract 1

2 The exact source statement 1

3 Definitions 2

4 The counterexample 2

5 Why the Leech-lattice lead does not decide the problem 3

6 Independent review and audit publication 3

6.1 Exact production timeline 4

7 Scope and significance 4

7.1 What is proved 4

7.2 What is not proved 4

8 External review checklist 5

References 5

Review status. The source mapping, counterexample, and atomic audit publication each passed an independent review. The audit root records this item as `independently_verified`. This is an open technical report, not a claim of journal peer review.

1 Abstract

AIM Problem 1.44 asks for a proof that no lattice has every shell a spherical 12-design. The official HTML, TeX, and PDF versions of the problem impose no dimension restriction. For any $a > 0$, take the full rank-one lattice $L = a\mathbb{Z} \subset \mathbb{R}$. Every nonempty positive-norm shell of L is a pair $\{-ak, ak\}$; after radial normalization it is exactly the full zero-sphere $S^0 = \{-1, 1\}$. Its uniform average therefore equals the spherical average for every polynomial, with no degree restriction. Thus every nonempty shell is a spherical t -design for every t , in particular for $t = 12$.

This gives a complete counterexample to the literal dimension-unqualified statement. It does not address the modified problem obtained by adding $n \geq 2$. Independent review also confirmed that the frequently cited Leech-lattice $11\frac{1}{2}$ -design property is not a 12-design witness: that notation deliberately skips degree 12, and the Leech minimum shell is explicitly not a 12-design.

2 The exact source statement

The official problem list for the AIM workshop *Discrete geometry and automorphic forms* prints the following item, attributed to Kumar:

Problem 1.44. “Prove there is no lattice whose shells are spherical 12-designs.”

The same one-sentence body occurs in the official HTML, TeX, and PDF exports. None of them states a lower bound on dimension or rank, or imposes integrality, parity, unimodularity, or self-duality.

The database label attached to the audited input had created a source-mapping trap: the number 98 in the AimPL aggregate data is the ordering position of the entire problem list, not a problem number. The target itself is Problem 1.44. A different Problem 1.32 appears on the same aggregate page but is mathematically unrelated. The audit recovered the complete item before deciding the claim.

3 Definitions

Let $L \subset \mathbb{R}^n$ be a full lattice. For $m > 0$, its norm- m shell is

$$L_m = \{x \in L : \langle x, x \rangle = m\}.$$

As usual in all-shell design statements, the property concerns nonempty positive-norm shells. If $L_m \neq \emptyset$, its radial normalization is the finite subset

$$X_m = m^{-1/2}L_m \subset S^{n-1}.$$

A nonempty finite set $X \subset S^{n-1}$ is a spherical t -design if, for every polynomial P of degree at most t ,

$$\frac{1}{|X|} \sum_{x \in X} P(x) = \int_{S^{n-1}} P(x) d\sigma(x),$$

where σ is normalized rotation-invariant measure. This is the standard averaging definition used in the lattice-design literature [2, 3].

In dimension one,

$$S^0 = \{-1, 1\}, \quad \int_{S^0} P(x) d\sigma(x) = \frac{P(-1) + P(1)}{2}.$$

No harmonic-polynomial equivalence theorem is needed below: the normalized shell will be the entire sphere itself.

4 The counterexample

Theorem 1 (Rank-one witness). *For every real $a > 0$, every nonempty positive-norm shell of the lattice $L = a\mathbb{Z} \subset \mathbb{R}$ is a spherical t -design for every nonnegative integer t . Consequently, L is a counterexample to the literal form of AIM Problem 1.44.*

Proof. Fix $m > 0$ with $L_m \neq \emptyset$. Then $m = a^2k^2$ for some integer $k \geq 1$, and the only lattice vectors of norm m are

$$L_m = \{-ak, ak\}.$$

Since $\sqrt{m} = ak$, radial normalization gives

$$X_m = m^{-1/2}L_m = \{-1, 1\} = S^0.$$

Therefore for every polynomial P , without any degree restriction,

$$\frac{1}{|X_m|} \sum_{x \in X_m} P(x) = \frac{P(-1) + P(1)}{2} = \int_{S^0} P(x) d\sigma(x).$$

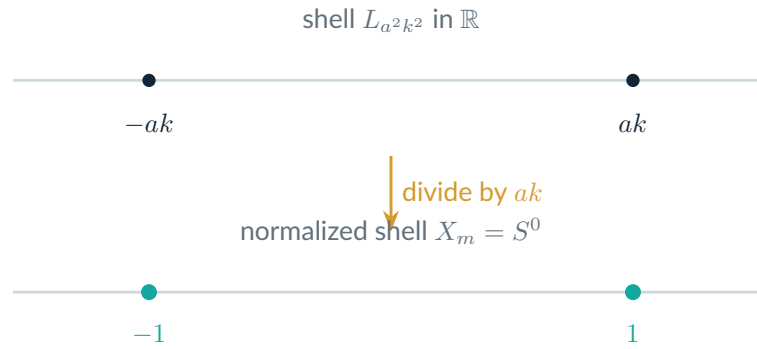


Figure 1: Every nonempty shell of $a\mathbb{Z}$ becomes the complete zero-sphere after radial normalization.

Thus X_m is a spherical t -design for every t , including $t = 12$. The argument applies to every nonempty positive-norm shell of $a\mathbb{Z}$. $\square$

The special choice $a = 1$ gives $L = \mathbb{Z}$, which is integral and self-dual. Thus even adding either of those two unstated adjectives would not remove the witness; an explicit exclusion of rank one would.

5 Why the Leech-lattice lead does not decide the problem

Pache proves that all shells of the Leech lattice are spherical $11\frac{1}{2}$ -designs [3]. This notation is not an ordinary strength between 11 and 12. For an antipodal set, an $11\frac{1}{2}$ -design satisfies the even harmonic conditions in degrees

$$2, 4, 6, 8, 10, 14,$$

whereas a spherical 12-design requires the conditions in degrees

$$2, 4, 6, 8, 10, 12.$$

The half-strength notation deliberately skips degree 12 and adds degree 14. Moreover, the proof immediately preceding Pache's Theorem 32 states that the Leech minimum shell cannot be a 12-design. The Leech lattice is therefore not an all-shell 12-design counterexample.

This correction matters because the audited input initially carried only the Leech $11\frac{1}{2}$ -design lead. The decisive argument instead comes directly from the unqualified dimension in the official AIM statement and the rank-one lattice $a\mathbb{Z}$.

6 Independent review and audit publication

The result passed two separate review gates:

1. An independent mathematical Reviewer recovered the full AIM item from the official HTML, TeX, PDF, and a historical page; checked that no dimension qualifier was hidden in the surrounding text; verified the S^0 averaging identity; and accepted the counterexample with the explicit $n \geq 2$ limitation.
2. A fresh integration Reviewer independently checked the atomic publication. The previous 297 audit rows remained byte-identical, the only new ID was 20001760, all scientific fields matched the accepted staging row, and only `review_status` was promoted to `independently_verified`.

The resulting audit root contains 298 unique reviewed IDs and has SHA-256

94a5d11151a1a877d4bc367c476a5b8bacb44244b4ee3d9001ac6a43d29639ea

6.1 Exact production timeline

All times below are UTC on 30 August 2026.

Time	Elapsed	Milestone
04:55:05	00:00:00	Source-recovery and rank-one-counterexample mission started.
05:21:39	00:26:34	Engineer completed the source map, proof, evidence package, and staging row.
05:28:27	00:33:22	Independent mathematical Reviewer accepted the literal counterexample.
05:33:06	00:38:01	Separate atomic-publication mission started.
05:41:18	00:46:13	The single-row atomic replacement completed with persisted transaction evidence.
05:50:06	00:55:01	Fresh integration Reviewer accepted the published 298-row root.

Thus the mathematical result reached independent acceptance in approximately **33 minutes**; the end-to-end path through independently reviewed publication took approximately **55 minutes**. Much of that time was source-identity recovery and audit verification. Once the unrestricted wording was established, the mathematical witness itself was immediate.

7 Scope and significance

7.1 What is proved

The exact sentence printed as AIM Problem 1.44 is false: a full rank-one lattice has every nonempty shell a spherical 12-design. The proof is exact, self-contained, and does not depend on numerical search.

7.2 What is not proved

- No conclusion is claimed for a repaired statement requiring dimension or rank at least two.
- No classification of higher-dimensional all-shell design lattices is claimed.
- The report does not claim that the likely intended higher-dimensional problem is easy or resolved.
- No worldwide-priority or novelty claim is made.

The result is best viewed as a precise formulation correction backed by a complete counterexample. Its natural scale is a short technical note: mathematically decisive for the literal wording, but deliberately limited in scope.

8 External review checklist

An external reviewer can check the complete argument in a few steps:

1. Read the complete official body of AIM Problem 1.44 and verify that it contains no dimension restriction.
2. Confirm that $a\mathbb{Z}$ is a full rank-one lattice in $\mathbb{R}$.
3. For every nonempty positive-norm shell, derive $L_{a^2k^2} = \{-ak, ak\}$.
4. Normalize by ak and verify that the resulting set is exactly $S^0 = \{-1, 1\}$.
5. Compare the uniform two-point average with normalized spherical measure on S^0 .
6. Keep the literal statement separate from a proposed $n \geq 2$ variant.
7. Do not substitute the Leech $11\frac{1}{2}$ -design property for a 12-design claim.

References

- [1] American Institute of Mathematics, *Discrete Geometry and Automorphic Forms: Problem 1.44*, official AimPL HTML, TeX, and PDF problem-list exports. <http://aimpl.org/discreteaf/>
- [2] Boris Venkov, notes by Jacques Martinet, “Réseaux et designs sphériques” [“Lattices and Spherical Designs”], Definition 3.1 and surrounding discussion.
- [3] Claude Pache, “Shells of Selfdual Lattices Viewed as Spherical Designs,” *International Journal of Algebra and Computation* **15** (2005), nos. 5–6, 1085–1127. <https://doi.org/10.1142/S0218196705002797>

Result 20001760 • Shareable report:

https://open.argusbot.cn/results/20001760/ARGUS_RESULT_20001760_TECHNICAL_REPORT.pdf

Prepared 30 August 2026 • Comments and independent checks are welcome.